**Esercitazione verifica: frazioni algebriche**

Calcolo C.E., semplificazione, moltiplicazione e divisione

**Soluzioni**

## Esercizio 1 – Condizioni di esistenza

**a)** $\dfrac{1}{3x}$

$$3x \neq 0 \quad \Longrightarrow \quad \text{C.E.} : x \neq 0$$

**b)** $\dfrac{1}{4x+20} = \dfrac{1}{4(x+5)}$

$$4(x+5) \neq 0 \quad \Longrightarrow \quad \text{C.E.} : x \neq -5$$

**c)** $\dfrac{1}{x^2+16}$

$$x^2 + 16 > 0 \quad \forall x \in \mathbb{R} \quad \Longrightarrow \quad \text{C.E.} : x \in \mathbb{R}$$

**d)** $\dfrac{x-1}{9-x^2} = \dfrac{x-1}{(3-x)(3+x)}$

$$(3-x)(3+x) \neq 0 \quad \Longrightarrow \quad \text{C.E.} : x \neq 3,\ x \neq -3$$

## Esercizio 2 – Semplificazione

**a)** $\dfrac{x^7+x^5}{x^2+1} = \dfrac{x^5(x^2+1)}{x^2+1}$

$$\text{C.E.} : x^2 + 1 > 0 \quad \forall x \in \mathbb{R}$$

$$\frac{x^5(x^2+1)}{x^2+1} = x^5$$

**b)** $\dfrac{x^2-3x+2}{2x-2} = \dfrac{(x-1)(x-2)}{2(x-1)}$

$$\text{C.E.} : 2(x-1) \neq 0 \quad \Longrightarrow \quad x \neq 1$$

$$\frac{(x-1)(x-2)}{2(x-1)} = \frac{x-2}{2}$$

**c)** $\dfrac{25-x^2}{x^2-10x+25} = \dfrac{(5-x)(5+x)}{(x-5)^2}$

$$\text{C.E.} : (x-5)^2 \neq 0 \quad \Longrightarrow \quad x \neq 5$$

$$\frac{(5-x)(5+x)}{(x-5)^2} = \frac{-(x-5)(x+5)}{(x-5)^2} = \frac{5+x}{5-x}$$

## Esercizio 3 – Moltiplicazione e divisione

**a)** $\dfrac{x^3}{x^2-4}\cdot\dfrac{x+2}{x^2}=\dfrac{x^3}{(x-2)(x+2)}\cdot\dfrac{x+2}{x^2}$

$$\text{C.E. : } (x-2)(x+2)\neq 0,\quad x^2\neq 0 \quad\Longrightarrow\quad x\neq -2,\ x\neq 0,\ x\neq 2$$

$$\frac{x^3(x+2)}{(x-2)(x+2)x^2}=\frac{x}{x-2}$$

**b)** $(4x^2-12x+9):(4x^2-9)=(2x-3)^2:\big[(2x-3)(2x+3)\big]$

$$\text{C.E. : } (2x-3)(2x+3)\neq 0 \quad\Longrightarrow\quad x\neq \frac{3}{2},\ x\neq -\frac{3}{2}$$

$$\begin{aligned}(2x-3)^2:\big[(2x-3)(2x+3)\big]&=\frac{(2x-3)^2}{1}\cdot\frac{1}{(2x-3)(2x+3)}\\&=\frac{(2x-3)^2}{(2x-3)(2x+3)}=\frac{2x-3}{2x+3}\end{aligned}$$